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French (native); English (fluent)
R (fluent)

Summary of publications

Total: 38 peer-reviewed publications of which 19 first/last/corresponding
+ 5 in BioRxiv (5 first/last/corresponding)
+ 4 yet unpublished thesis chapters (4 first/last/corresponding)

Google scholar profile: <https://scholar.google.fr/citations?user=VhsB4z0AAAAJ&hl=en>

- H-index: 22
- Total citations: ~1950 (31 publications with more than 10 citations)

Complete publication record

Below I have listed my publications from the most recent to the oldest in the following categories:

- Publications with a leading role
- Publications with a supporting role (where I indicate my contribution)
- Publications in bioRxiv
- Thesis chapters or exist as draft (as they reflect future publications)

Publications with a leading role († corresponding author; * equal contribution)2021 to date:

1. Michalak P, J-B Ferdy†*, Duneau D†* *The role of fecundity dynamics and time of injury for female reproductive fitness*. **Roy. Soc. Open Science**

We studied experimentally the impact of wounds on reproduction and tested the hypothesis that injury at different periods of the fecundity dynamics will have different impact on fitness. We found that females were able to quickly compensate for the short-term consequences of wounding.

2. Michalak P, Duneau D†*, Ferdy J-B†*. *Seminal fluid proteins as regulation factors for optimizing reproduction: a modeling approach*. **American Naturalist** doi.org/10.1086/740809

Developed a biologically informed theoretical model showing how seminal fluid proteins synchronize sperm and egg release, reducing unfertilized egg production. Identified SP exhaustion as the key synchronization signal and demonstrated that sexual conflict over SP-mediated regulation of female physiology should be limited, though conflict over optimal remating rate remains possible.

3. Jerónimo M, Garizo A, Atencio G, Duneau D†*, Beldade P†*. (2025) *Wound-induced eyespots on butterfly wings at the intersection of immune response and pigmentation development* **BMC Biology** doi.org/10.1186/s12915-025-02397-3

Showed that immune activation modulates wound-induced ectopic eyespot formation in butterfly wings, revealing crosstalk between immune function and pigmentation development.

4. Duneau D†*, Lafont P*, Lauzeral P, Parthuisot N, Faucher C, Xuerong J, Buchon N, Ferdy JB†. (2025) *A within-host infection model to explore tolerance and resistance*. **eLife** doi.org/10.7554/eLife.104052

Developed a within-host infection model integrating theory and experiments to distinguish between tolerance and resistance mechanisms. Provided a practical experimental framework for explaining individual differences in susceptibility to infection, a methodological foundation applied in subsequent work (e.g. Monteiro-Black et al. 2025).

5. Lafuente E, Duneau D, Beldade P. (2024) *Genetic architecture of plasticity for pigmentation components in Drosophila melanogaster*. **Molecular Ecology** doi.org/10.1111/mec.17294

Applied genome-wide association mapping in *Drosophila melanogaster* to dissect the genetic basis of temperature-dependent pigmentation plasticity. Found that different body parts rely on largely independent genetic architectures, challenging the assumption of a common genetic basis for pigmentation plasticity. Contribution: I supervised the student for the genomic analysis and for the validation of allele candidate with functional genetics.

6. **Duneau[†] D**, Möst M, Ebert D. (2022) *Evolution of sperm morphology in Daphnia species*. **Peer Community Journal** (Diamond Scientific Publishing) doi.org/10.24072/pcjournal.182

Based on a recent phylogeny, we identified that *Daphnia* had among the smallest recorded sperm and studied the evolution of sperm length in this clade.

7. **Duneau[†] D**, Buchon N (2022) *Gut cancer increases the risk for Drosophila to be preyed upon by hunting spiders*. Accepted in **Animal Behaviour** doi.org/10.1016/j.anbehav.2022.07.003

Diseased individuals are more preyed upon than healthy individuals. Although this is intuitive, this conventional wisdom has been subjected more to speculation than to empirical study. To test the idea, we genetically induced colon cancer in *Drosophila* and found that cancerous flies are more frequently predated than healthy ones.

8. **Duneau[†] D**, Ferdy J-B (2022) *Pathogen within-host dynamics and disease outcome: what can we learn from insect studies?* Accepted in **Current Opinion in Insect Science** doi.org/10.1016/j.cois.2022.100925

I have been invited by Sophie Armitage (University of Berlin) and Barbara Milutinović (University of Münster) to review the latest knowledge on the connection between within-host pathogen dynamics and disease outcome.

9. Rodrigues YK, van Bergen E, Alves F, **Duneau D***, Beldade P* (2021) *Additive and non-additive effects of day and night temperatures on thermally plastic traits in a model for adaptive seasonal plasticity*. **Evolution** evo.14271

We tested the effects of circadian temperature fluctuations on a series of thermal plasticity traits in a model of adaptive seasonal plasticity, the *Bicyclus anynana* butterfly.

10. Faucher C*, Mazana V*, Kardacz M, Parthuisot N, Ferdy J-B, **Duneau[†] D** (2021) *Step-specific adaptation and trade-off over the course of an infection by GASP-mutation small colony variants*. **mBio** doi.org/10.1128/mBio.01399-20

Within-host bacterial adaptations are generally focused on antibiotic resistance, rarely on the adaptation to the environment given by the host, and the potential trade-off hindering adaptations to each step of the infection are rarely considered. Using *Drosophila melanogaster* as host and the bacteria *Xenorhabdus nematophila*, we studied those trade-offs that are key to understand intra-host evolution, and thus the dynamics of the infection. We found that adaptation to late infection trade-off with adaptation to early infection.

2020 and before:

11. Belmonte RL, Corbally M-K, **Duneau D***, Regan JC*[†] (2020) *Sexual dimorphisms in innate immunity and responses to infection in Drosophila melanogaster*. **Frontiers in Immunology** doi.org/10.3389/fimmu.2019.03075

We evaluated the sexual dimorphism of *Drosophila* immune responses based on an exhaustive survey of studies on the immunity of this model. We highlighted that current theories are too generalist to have sufficient explanatory power, and so, generated a set of realistic, testable, hypotheses from which my project derives.

12. **Duneau D***, Sun H*, Revah J, San Miguel K, Kunerth HD, Caldas IV, Messer PW, Scott JG, Buchon N (2018) *Genome wide analysis of resistance to an organophosphate and a pyrethroid insecticide*. **G3: Genes/Genomes/Genetics** doi.org/10.1534/g3.118.200537

Using GWAS with the *Drosophila* Reference Genetic Panel (DGRP) found the genetic basis of the resistance to Parathion and Deltamethrin, two commonly used insecticides. We showed that commonly used insecticides have a strong impact on the evolution of non-target species such as *Drosophila* and that resistance alleles are present worldwide.

13. Lafuente E, **Duneau D**, Beldade P (2018) *Genetic basis of thermal plasticity variation in Drosophila melanogaster body size*. **PLoS Genetics** doi.org/10.1371/journal.pgen.1007686

Using GWAS in *Drosophila*, we determined the genetic basis of thermal plasticity of thorax and abdomen size. Contribution: I supervised the experimental design, the genomic analysis and the validation of allele candidates with functional genetics.

14. **Duneau[†] D**, Lazzaro B (2018) *Persistence of an extracellular systemic infection across metamorphosis in a holometabolous insect*. **Biology Letters** doi.org/10.1098/rsbl.2017.0771

We showed that systemic infection with an extracellular bacterium can transverse life stages.

15. **Duneau[†] D**, Ferdy JB, Revah J, Kondolf HC, Ortiz GA, Lazzaro BP, Buchon N (2017) *Stochastic variation in the initial phase of bacterial infection predicts the probability of survival in D. melanogaster*. **eLife** doi.org/10.7554/eLife.28298 ()

A central problem in biomedicine is understanding why individuals exposed to seemingly identical infections can have radically different outcomes. Using *Drosophila melanogaster*, we combined functional genetics and mathematical

modeling to analyze the main determinants of stochastic infection outcome. We showed that even without genetic variance and in a controlled environment, some hosts survive while others succumb. We characterised two new bacterial load measures (load upon death and load during chronicity) to describe the infectious process, and demonstrated that host survival is predicted by within-host bacterial proliferation, with small variations in innate immune response differentiating survivors from non-survivors. We developed an integrative dynamic model that accurately predicts infection outcome and shows that the timing of immune response initiation determines survival. My research program is largely built on the knowledge and methods acquired during this study. ("Highlight" by Graham and Tate eLife 2017 6:e32783; recommended 4× "very good" on Faculty Opinions, <https://archive.connect.h1.co/article/731979509>)

16. **Duneau[†] D**, Kondolf HC, Im JH, Ortiz GA, Chow C, Fox MA, Eugénio AT, Buchon N, Lazzaro BP (2017) *The Toll pathway underlies host sexual dimorphism in resistance to both Gram-negative and positive-bacteria in Drosophila*. **BMC Biology** doi.org/10.1186/s12915-017-0466-3

We elucidated part of the mechanisms underlying the difference between sexes in infection outcome in *Drosophila*. We revealed that the Toll pathway, responsible for the sexual dimorphism in some human infectious diseases (e.g. HIV), also explains why female *Drosophila* are slower than males to control pathogen proliferation.

17. **Duneau[†] D**, Ebert D, Du Pasquier L (2016) *Infections by Pasteuria do not protect its natural host Daphnia magna from subsequent infections*. **Developmental & Comparative Immunology** doi.org/10.1016/j.dci.2015.12.004

We tested and concluded that *Daphnia* do not have immunological memory upon bacterial infections.

18. **Duneau[†] D**, Ebert D (2012) *Host sexual dimorphism and parasite adaptation*. **PLoS Biology** doi.org/10.1371/journal.pbio.1001271

In this "essay" we propose for the first time the idea that the sexual dimorphism of diseases may be the result of the specific adaptation of parasites to the sex of their host. Similarly, as organisms adapt to the environment to which they are most frequently exposed, parasites can adapt to the sex they encounter most frequently (e.g., either because males and females are exposed differently, or because one sex is more easily infected than another due to immune differences). As a result, parasites behave differently depending on the sex they infect.

19. **Duneau[†] D**, Luijckx P, Ruder L, Ebert D (2012) *Sex-specific effects of a parasite evolving in a female-biased host population*. **BMC Biology** doi.org/10.1186/1741-7007-10-104

We have shown here that a parasite can be better adapted to the sex it encounters most frequently, empirically supporting the hypothesis proposed Duneau & Ebert PLoS Biology 2012.

20. **Duneau[†] D**, Ebert D (2012) *The role of molting in parasite defense*. **Proceedings of the Royal Society of London B** doi.org/10.1098/rspb.2012.0407

We show that moulting is not only a weakness but can be beneficial to prevent infection by shedding bacteria. (Covered by: Le Figaro (<https://bit.ly/2tbIPUK>) & Live Science (<https://bit.ly/37aT6Po>))

21. **Duneau[†] D**, Luijckx P, Ben-Ami F, Laforsch C, Ebert D (2011) *Resolving the infection process reveals striking differences in the contribution of environment, genetics and phylogeny to host-parasite interactions*. **BMC Biology** doi.org/10.1186/1741-7007-9-11

We investigated the mechanism of infection underlying coevolution between a host (*Daphnia magna*) and his parasite (*Pasteuria ramosa*). We found that the specificity depends on the capacity of the parasite to attach or not to the host oesophagus. I published here the "attachment test" method which is still used routinely in my PhD supervisor lab to quickly determine the ability of the bacteria of a given genotype to infect a given host genotype. It is the first study to describe a mechanism responsible for the coevolution between an animal host and its parasite.

22. Ponton* F, **Duneau* D**, Sanchez M, Courtiol A, Terekhin A, Budilova EV, Renaud F, Thomas F (2009) *Effect of parasite-induced behavioral alterations on juvenile development*. **Behavioral Ecology** doi.org/10.1093/beheco/arp092

We studied whether the juveniles of manipulated female *Gammarus* (i.e. with a high risk of predation) developed more rapidly in their mother's ventral pouch, a behaviour that could have been selected to reduce the juveniles' chances of predation; this is not the case.

23. **Duneau D**, Bouludier T, Gomez-Diaz E, Petersen A, Tveraa T, Barrett RT, McCoy KD (2008) *Prevalence and diversity of Lyme borreliosis bacteria in marine birds*. **Infection, Genetics and Evolution** doi.org/10.1016/j.meegid.2008.02.006

We studied the diversity of *Borrelia* spp. circulating in seabirds. Our findings indicate that seabirds may be an important component in the global epidemiology and evolution of Lyme disease.

Publications with a supporting role

2021 to date:

24. Bonfini A, Dobson AJ, **Duneau D**, Revah J, Liu X, Houtz P, Buchon N (2021) *Diet composition plastically resizes the Drosophila midgut by affecting cell gain and loss, stem cell-niche coupling and enterocyte size.* **eLife** doi.org/10.7554/eLife.64125

We studied the phenotypic plasticity of *Drosophila* gut in response to glucose level in diet. Contribution: Performed the GWAS and all the statistical analyses of the paper.

2020 and before:

25. Bento G, Fields P, **Duneau D**, Ebert D (2020) *An alternative route of bacterial infection is associated with a polymorphism at an alternative resistance locus.* **Heredity** doi.org/10.1038/s41437020 -0332-x

Contribution: I discovered the alternative route of infection.

26. Pineaux M, Merklings T, Danchin E, Hatch S, **Duneau D**, Blanchard P, Leclaire S (2020) *Sex and hatching order modulate the association between MHC-diversity and fitness in early-life stages of a wild seabird.* **Molecular Ecology** doi.org/10.1111/mec.15551

Contribution: Helped in the analyses and writing as an evolutionary parasitologist knowing the immune system and the sexual dimorphism in diseases.

27. Corse E, Tougaard C, Archambaud G, Agnès J-F, Messu Mandeng FD, Bilong Bilong CF, **Duneau D**, Zinger L, Chappaz R, Xu CCY, Mègelecz E, Dubut V (2019) *One-locus-several-primers: a strategy to improve the taxonomic and haplotypic coverage in diet metabarcoding studies.* **Ecology & Evolution** doi.org/10.1002/ece3.5063

Contribution: Sampled spider webs in the tropical rainforest of French Guyana to show that we used them as DNA traps to describe biodiversity with metabarcoding.

28. Ebert D, **Duneau D**, Hall M, Luijckx P, Andras J, Du Pasquier L, Ben-Ami F (2016) *A population biology perspective on the stepwise infection process of the bacterial pathogen Pasteuria ramosa in Daphnia.* **Advances in parasitology** doi.org/10.1016/bs.apar.2015.10.001

We demonstrate that a population biology approach taking into consideration the natural genetic and environmental variation at each step of the infection can greatly aid our understanding of the evolutionary processes shaping disease traits. Contribution: This manuscript is largely based on my PhD thesis.

29. Avila F, Cohen A, Ameerudeen F, **Duneau D**, Suresh S, Mattei A, Wolfner M (2015) *The Drosophila mating plug protein, PEBme, is required to maintain the ejaculate within the female reproductive tract at the termination of copulation.* **Genetics** doi.org/10.1534/genetics.115. 176669

Contribution: Showed with macrophotography that the inability of females to subsequently retain the ejaculate in their reproductive tracts after mating was frequent.

30. Luijckx P, **Duneau D**, Andras J, Ebert D (2014) *Cross-species infection trials reveal cryptic parasite varieties and a putative polymorphism shared among host species.* **Evolution** doi.org/10.1111/evo.12289

Contribution: Designed and did most experiments together with the first author.

31. Luijckx P, Fienberg H, **Duneau D**, Ebert D (2013) *A matching-allele model explains host resistance to parasites.* **Current Biology** doi.org/10.1016/j.cub.2013.04.064 ()

First times the genetic model of coevolution, the matching-allele model, was shown. Contribution: Performed all the attachment tests of this study (based on Duneau et al. BMC Biol. 2011).

32. Luijckx P, Fienberg H, **Duneau D**, Ebert D (2011) *Resistance to a bacterial parasite in the crustacean Daphnia magna shows Mendelian segregation with dominance.* **Heredity** doi.org/10.1038/hdy.2011.122

Contribution: Performed all the attachment tests of this study (based on Duneau et al. BMC Biol. 2011).

33. Ponton F, Otalora-Luna F, Lefevre T, Guerin PM, Lebarbenchon C, **Duneau D**, Biron DG, Thomas F (2011) *Water-seeking behavior in worm-infected crickets and reversibility of parasitic manipulation.* **Behavioral Ecology** doi.org/10.1093/beheco/arq215

Contribution: Participated in sampling and performed several experiments of the study.

34. Gómez-Díaz E, Doherty P Jr, **Duneau D**, McCoy KD (2010) *Cryptic vector divergence masks vector-specific patterns of infection: an example from the marine cycle of Lyme borreliosis*. **Evolutionary Applications** doi.org/10.1111/j.1752-4571.2010.00127.x

Contribution: Performed everything based on Sanger PCR.

35. McCoy KD, **Duneau D**, Bouludier T (2008) *Spécialisation de la tique des oiseaux marins et diversité des bactéries du complexe Borrelia burgdorferi sensu lato, agents de la maladie de Lyme : effets en cascade dans les systèmes à vecteur*. **Les actes du BRG 277-291** (french publication with reviewing committee)

Summary of the work in Duneau et al IGE 2008 for the colloquium BRG.

36. Ponton F, Lebarbenchon C, Lefèvre T, Biron DG, **Duneau D**, Hughes DP, Thomas F (2006) *How parasitic Gordian worms cut the Gordian knot: a novel solution to predation upon the host*. **Nature** doi.org/10.1038/440756a

As prisoners in their living habitat, parasites should be vulnerable to the predators of their hosts. We show that the parasitic hairworm can escape from the predators of its hosts. Contribution: Performed all the experiments with frogs (one of the three predators).

37. Ponton F, Lebarbenchon C, Lefèvre T, Thomas F, **Duneau D**, Marché L, Renault L, Hughes DP, Biron DG (2006) *Hairworm anti-predator strategy: a study of causes and consequences*. **Parasitology** doi.org/10.1017/S0031182006000904

Contribution: I designed the whole nursery for the hairworms and monitored all the mating to measure the fitness.

38. Ponton F, Biron DG, Joly C, **Duneau D**, Thomas F (2005) *Ecology of populations parasitically modified: a case study from a gammarid (Gammarus insensibilis)-trematode (Microphallus papillorobustus) system*. **Marine Ecology-Progress Series** doi.org/10.3354/meps299205

We studied the consequence of behavioural manipulation by parasites on the ecology of host populations. Contribution: I was involved in most of the experiments and quantified all host energy resources (glucose, lipid, etc.).

Publications in bioRxiv († corresponding author; * equal contribution)

39. Michalak P, Duneau D†*, J-B Ferdy†* *Seminal fluid proteins can mitigate sexual conflict: the case of remating in insects*. doi.org/10.64898/2026.02.27.708467 (Submitted to **PNAS**)

We used a mathematical model of reproductive physiology, informed by accumulated knowledge on *Drosophila melanogaster*, to examine the role of seminal fluid proteins in sexual conflict over optimal remating rate. Our results suggest that while a conflict over the remating rate does exist, seminal fluid proteins reduce its intensity, highlighting their role in aligning the interests of both sexes.

40. Rubinic M, Arias-Rojas A, Martinez K, Klimek W, Paczia N, Alagesan K, Duneau D, Iatsenko I *Sex differences in Drosophila intestinal metabolism contribute to sexually dimorphic infection outcome and alter gut pathogen virulence*. doi.org/10.1101/2025.05.22.655590 (Second revision for **PNAS**)

We studied the sexual dimorphism of intestinal infections of *D. melanogaster*. We showed that males exhibit increased a key antioxidant defence system. It allows them to withstand oxidative stress-induced defecation blockage and clear the pathogen from the intestine, resulting in survival. We also showed that the bacteria showed increased expression of several virulence factors in female gut, indicating a change in pathogen behaviour depending on the host they infect (a theory I have been developing since Duneau & Ebert PLoS Biol. 2012). I have participated in developing the hypothesis, trained the first author to perform the Bulk RNAseq analysis and in using R. I also analysed the GWAS.

41. Monteiro-Black B, Corbally MK, Aleksandrowicz J, Taggart AM, Langella F, Belmonte B, Mika P, **Duneau D**†*, Regan J†* *Different ways to die: harnessing variation in Drosophila reveals loss of tolerance and resistance over age and sex-specific association between infection susceptibility and lifespan*. doi.org/10.1101/2025.07.07.663438

Using methods developed in Duneau et al. eLife 2017 and Duneau et al. eLife 2025, we showed that immune ageing can underlie a decrease in the ability to sustain pathogens, not only to control their proliferation.

42. **Duneau^{†*} D**, Gallet R^{*}, Adhiambo M, Delétré E, Chailleux A, Khamis F, Subramanian S, Brévault T, Fellous S[†] *Sex-specific sub-lethal effects of low virulence entomopathogenic fungi may boost the Sterile Insect Technique*. doi.org/10.1101/2024.03.07.583916

Our experimental study supports a paradigm shift in pest control from the "maximally virulent" approach to a "sex-dependent" strategy aimed at selecting bioagents for boosted Sterile Insect Techniques. Such an approach can maintain the competitiveness of sterile males to access the females, while contributing to horizontal transmission of pathogens which increases female mortality and decrease their fertility.

43. **Duneau[†] D**, Altermatt F, Ferdy J-B, Ben-Ami F, Ebert D. *Estimation of the propensity for sexual selection in a cyclical parthenogen*. doi.org/10.1101/2020.02.05.935148

Our data from the field show that sexual selection is present in *Daphnia*, although it is mainly parthenogenetic, and that this selection probably manifests itself through a combination of female choice and male competition.

Thesis chapters or exist as draft

1. Kambou S, Liu C, Tam-Hui T, Gineste A, **Duneau D^{†*}**, Chassagne F^{†*} *Using the *Drosophila/Serratia marcescens* system to test in vivo new antimicrobial compounds*. (First revision for **Applied and Environmental Microbiology**)

We used my expertise in *Drosophila* infections and the expertise of François Chassagne in ethnopharmacology to successfully develop a method of *in vivo* screening of antimicrobial compounds present in traditional medicine.

2. Rodrigues YK, **Duneau D^{†*}**, Beldade P^{†*}. *Seasonal and sexual dimorphism in immunity in a thermal plasticity model*.

Using methods developed in Duneau et al. eLife 2017 with *Drosophila*, we studied the thermal phenotypic plasticity of the immune system of the butterfly *Bicyclus anynana*.

3. Corbally MK, **Duneau D^{*}**, Regan J^{*} *Investigating the role of a putatively longevity-associated transposable element in lifespan and immunity*.

Using GWAS with transposable element (TE), instead of SNP, in *Drosophila*, we discovered that a TE inclusion in regulatory region of major immune gene affected lifespan and the response to infection.

4. Xiao-Li Bing^{*}, Peter Nagy^{*}, **Duneau D^{*}**, Alessandro Bonfini, Katia Troha, Nicolas Buchon. *Deciphering similarities and differences between *Drosophila melanogaster* and *Drosophila suzukii* in response to bacterial infections*.

We studied the response of infection between a fruit pest *Drosophila* (*D. suzukii*) and the reference model organism *D. melanogaster*. We showed a surprising similarity, suggesting that a large part of the knowledge on *D. melanogaster* could be used to fight a very important crop pest. I have been involved in the analysis of the bacterial within-host dynamics and in the interspecies comparison of the transcriptomic response, which brings overlooked technical challenges.